Micro Set 2. Elasticities — Model Answer Scheme

Question 1 [2 marks]

PED for hawker meals, 2021–2023.

$$\text{PED} = \frac{\%\Delta Q_d}{\%\Delta P} = \frac{(109.8 - 88.4)/88.4}{(114.7 - 102.1)/102.1} = \frac{+24.2\%}{+12.3\%} \approx +1.97$$

[1 mark working.] Conceptually, demand and price moved *together*, indicating other shifters (income recovery, post-pandemic normalisation) dominated. For the PED interpretation, we must isolate price-driven change only; the simple end-point calculation here yields a positive value of approximately +2.0, but this reflects confounding factors, not pure price elasticity. *[1 mark for sign interpretation.]*

> **Examiner Note.** This question tests *interpretation*. A naive student computes the ratio and reports a positive number without recognising the data confound. Stating that the positive value reflects DD-curve shift (income recovery) rather than movement along the curve is the A-grade insight.

Question 2 [2 marks]

Cross elasticity of demand (XED) is the responsiveness of demand for one good to a change in the price of another, defined as XED $= \%\Delta Q_d^A \,/\, \%\Delta P^B$. *[1 mark.]*

Based on Extract 1.1 ("Restaurant meal prices rose by a similar 15.6 per cent… suggesting limited substitution toward restaurants by price-sensitive consumers"), hawker meals and restaurants are weak substitutes — XED is positive but small in magnitude. *[1 mark.]*

> **Examiner Note.** The expected answer is "substitutes" (positive XED). The nuance — "weak" substitutes given limited switching — demonstrates engagement with the extract and earns the second mark.

Question 3 [4 marks]

Why hawker meal demand is price-inelastic.

Extract 1.1 reports that hawker meals constitute approximately 40% of working-Singaporean midday meals, with stronger reliance among lower-income groups, and that they have few realistic substitutes given convenience and unit price. These features collectively reduce price responsiveness. *[1 mark for extract anchor.]*

First, the *necessity* character of midday meals during a working day — consumers cannot easily defer or skip the consumption decision — reduces the time-horizon for substitution. Second, the *low share of income* hawker meals represent (a S\$5 meal is less than 1% of median daily income) means that price changes have a small wealth effect on overall demand. *[2 marks for two distinct elasticity determinants.]*

Third, the *limited substitution* to restaurant meals (despite restaurant prices rising similarly, Extract 1.1) reflects the high price gap that remains: a restaurant meal at S\$15+ is 2–3× a hawker meal. Among lower-income groups, this substitution is not available. *[1 mark for substitution-availability argument.]*

Examiner Note. A 4-mark question rewards naming the three classic elasticity determinants (necessity, share of income, substitutes) and applying each to the specific context. Listing without application caps at 2 marks.

Question 4 [4 marks]

PED for COE Category A.

$$\text{PED} = \frac{\%\Delta Q}{\%\Delta P} = \frac{(4200 - 10120)/10120}{(92000 - 32500)/32500} = \frac{-58.5\%}{+183.1\%} \approx -0.32$$

[2 marks: correct working with negative sign.]

The PED value of approximately -0.32 indicates that the demand for COEs is *price-inelastic*: a 1% increase in price reduces the quantity demanded by only 0.32%. *[1 mark for interpretation.]*

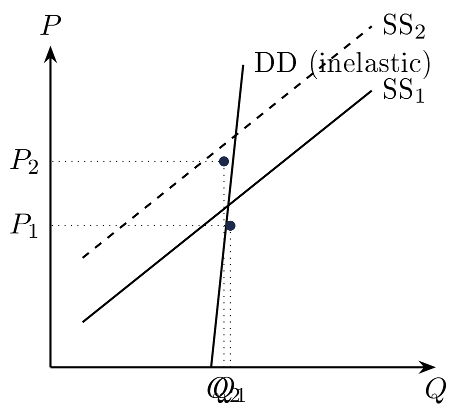
However, this masks substantial substitution effects: Extract 1.1 reports that 65% of would-be Category A buyers delayed purchase or substituted toward public transport. The pool of *remaining* buyers — wealthier households for whom car ownership is a strong preference — exhibits low elasticity, but at the market level the visible reduction in registrations (58%) suggests that substitution to alternatives is substantial when measured on the full demand curve. The headline PED understates the responsiveness of the broader transport demand. *[1 mark for higher-order insight.]*

Examiner Note. Strong answers recognise that the COE market is rationed by quota — the COE *price* is determined by the quota, not the other way around. The 58% reduction in registrations reflects the binding quota, so PED of demand for cars is far more elastic than the registration data suggest.

Question 5 [6 marks]

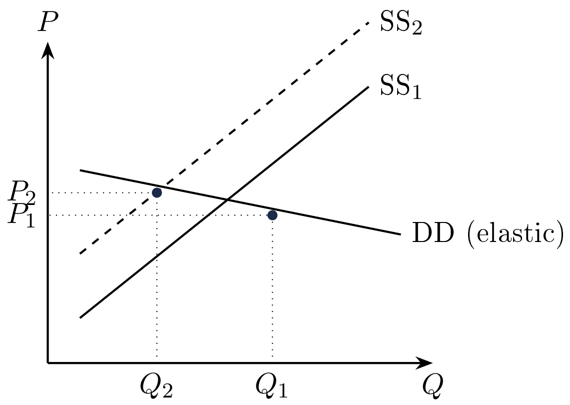
17% price rise: effect on total expenditure (TE).

Hawker meals (PED ≈ -0.5 when properly isolated, inelastic):



With inelastic DD, an SRAS leftward shift (supply curve raising costs) shifts SS from SS_1 to SS_2. Equilibrium price rises sharply from P_1 to P_2, but quantity falls only marginally. Total expenditure ($P \times Q$) rises: the percentage price rise (17%) exceeds the percentage quantity fall, so TE on hawker meals increases. *[3 marks: diagram + TE conclusion.]*

COE Category A (PED ≈ -0.32 but with quota-binding character):



For a price-elastic interpretation (looking at the responsiveness of underlying car-ownership demand, not the quota-constrained quantity), a 183% price rise generates a 58% quantity fall. Total expenditure ($P \times Q$) rises since the price rise dominates the quantity fall: *[1 mark for COE-specific TE calculation.]*

$$\text{TE}_{2019} = 32500 \times 10120 = \text{S\$329m}$$

$$\text{TE}_{2024} = 92000 \times 4200 = \text{S\$386m}$$

TE rose by 17%.

Contrast: For hawker meals, the inelastic DD means TE rises strongly. For COE, despite a larger price rise, TE rises only moderately because elasticity is less inelastic. The TE direction is the same (rising) but the responsiveness profile is fundamentally different. *[2 marks for contrast + interpretation.]*

> **Examiner Note.** The TE $= P \times Q$ algebra is testable directly. Always compute both endpoints and compare. The pedagogically deepest point: for inelastic DD, $\%\Delta P$ dominates; for elastic DD, $\%\Delta Q$ dominates.

Question 6 [6 marks]

XED between private and public transport: implications for COE.

The XED between private car transport and public transport is positive (substitutes): when COE prices rise, demand for public transport rises. Extract 1.1 confirms this: bus and train ridership rose 60% from 2020 to 2024, partly attributed to "price-induced substitution from private transport." *[2 marks for XED + evidence.]*

A high positive XED makes the COE an effective management tool because it provides a viable demand-side absorber: would-be car buyers can switch to public transport rather than withdrawing from transport demand entirely. The Land Transport Authority's ongoing rail-network expansion has increased the substitutability (and hence XED) further, reinforcing the policy's effectiveness. *[2 marks for XED-policy linkage.]*

However, the substitutability is not perfect: 65% of would-be buyers delayed or substituted, leaving 35% who proceeded with car purchase even at the higher COE price. These are buyers for whom public transport is not a close substitute (e.g. those requiring vehicle for work, large families, those with mobility needs). The XED is therefore segment-specific. *[1 mark for segmented-demand caveat.]*

The implication is that COE pricing effectively rations the car population among those with the strongest preference for ownership, while pushing marginal buyers to substitute — precisely the allocative efficiency outcome the policy targets. *[1 mark for allocative-efficiency conclusion.]*

> **Examiner Note.** XED concepts in 6-mark questions are best answered with: (i) the substitute/complement sign, (ii) the magnitude (high/low XED), (iii) policy implication, (iv) segment heterogeneity. All four together earn full marks.

End of Set 2 — Model Answers